



Society of Petroleum Engineers

SPE-226931-MS

Mixed Wettability Microfluidics: Fabrication, Experimental Validation, and Numerical Simulation

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226931-MS](https://doi.org/10.2118/226931-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Subsurface reservoirs often exhibit heterogeneous wettability patterns, named mixed wettability, which critically influence multiphase fluid flow dynamics and trapping mechanisms. To investigate these effects at the pore scale, microfluidic systems have become crucial. Despite their utility, the fabrication of microfluidic devices with precisely controlled spatial distributions of mixed wettability remains a significant challenge. This study presents a novel fabrication approach that combines photolithography and molecular vapor deposition (MVD) of perfluorodecyltrichlorosilane (FDTs) to engineer micromodels with defined mixed wettability configurations. Two micromodel designs were developed to emulate the wettability complexities of natural porous media, including single channels with horizontal and vertical wettability transitions. Two-phase flow experiments were conducted using high-resolution microscopy and high-speed imaging to capture dynamic pore-scale phenomena. Experimental observations highlighted the profound effect of mixed wettability on the behavior of two-phase fluids. Specifically, the meniscus shape shifted from convex to concave as fluids moved between hydrophobic and hydrophilic areas. In channels with vertical variations in wettability, the fluid interface formed a distinctive S-shape. These observations were complemented by numerical simulations using the phase-field method in COMSOL Multiphysics®, enabling validation of the experimental results. The simulations closely matched experimental findings, highlighting the efficiency of the fabrication technique and the reliability of the experimental and numerical methods. This work provides a robust platform for studying the interplay between mixed wettability and multiphase flow, offering valuable insights for applications in subsurface energy systems, including oil recovery, CO₂ sequestration, and hydrogen storage.

Introduction

Understanding pore-scale processes is crucial for optimizing energy and environmental applications such as hydrocarbon recovery, CO₂ sequestration, and H₂ storage (Heinemann et al., 2021; Metz et al., 2005; Tabrizinejadas et al., 2023; Thomas, 2008). For example, manipulating interfacial forces within porous media is crucial for optimizing oil recovery and enhancing CO₂ storage in deep saline aquifers (Valvatne and Blunt, 2004). These microscopic interactions not only influence multiphase flow at the pore scale

but also determine macroscopic performance, ultimately affecting the feasibility and success of field-scale operations.

Central to these applications is the process of fluid-fluid displacement, where one fluid replaces another in a confined porous space. This displacement is governed by the interaction of capillary and viscous forces, which dictate flow dynamics at the pore scale (Lake et al., 2014). A critical factor influencing this process is the wettability of the solid surface, which represents the tendency of a surface to preferentially interact with one fluid over another. Wettability significantly impacts two-phase flow behavior, determining the efficiency of fluid mobilization and displacement (AlOmier et al., 2024b; Blunt, 2017; de Gennes et al., 2004).

Subsurface reservoirs often exhibit mixed wettability, where hydrophobic and hydrophilic regions coexist within the porous structure. This heterogeneity, caused by mineralogical variations or the deposition of organic components like asphaltenes from oil, introduces complexities in fluid displacement and capillary trapping phenomena (Kovscek et al., 1993; Radke et al., 1992; Salathiel, 1973). The complex interaction of interfacial forces and flow pathways in mixed wettability systems poses considerable challenges for experimental studies and computational models, which remain less developed compared to those for homogeneous wettability scenarios. Core flooding experiments have been widely employed to study the influence of mixed wettability on multiphase flow in porous media. These experiments typically use either naturally sourced core samples, which retain their original reservoir conditions, or synthetic cores that are chemically treated and aged to simulate the mixed-wettability characteristics of reservoir rocks (Anderson, 1986). Observations from these studies reveal significant variations in flow behavior between mixed-wet and uniformly wet rocks (Alhammadi et al., 2020; Al-Menhali et al., 2016; Al-Menhali and Krevor, 2016; Lin et al., 2019; Scanziani et al., 2020; Zou et al., 2018). While core flooding experiments provide valuable insights into reservoir conditions, they have notable limitations. The opacity of core samples restricts direct observation of pore-scale mechanisms, and imaging techniques like CT are constrained by resolution and time. Consequently, key properties are often inferred indirectly, while structural complexity and wettability variations hinder accurate interpretation of flow behavior.

Microfluidic systems offer a promising alternative to overcome the limitations of traditional methods by enabling rapid, precise, and real-time analyses while replicating representative pore geometries. These systems have provided valuable insights into fluid displacement; however, most studies have focused on homogeneous wettability effects (Cottin et al., 2011; Hu et al., 2017; Lei et al., 2022; Zhao et al., 2016), largely neglecting the impact of mixed wettability. This gap stems from the challenges of fabricating microfluidic devices with controlled mixed-wet properties. Only a few efforts have addressed this issue, aiming to develop mixed-wet microfluidic devices (Chang et al., 2020; Irannezhad et al., 2023; Lee et al., 2015; Sun and Torres-Verdín, 2023). Yet, achieving precise spatial control of wettability distribution at high resolution remains a significant challenge. Recently, AlOmier et al. proposed an innovative method for selective wettability modification in microfluidic devices (AlOmier et al., 2024a). This approach employs photolithography and thin-film deposition of perfluorodecyltrichlorosilane (FDTS) to achieve precise, spatially distributed mixed-wet surfaces, providing a high level of control and flexibility in wettability patterns.

The objective of this work is to validate the proposed technique for fabricating microfluidic devices with controlled mixed-wettability configurations and to explore their impact on two-phase flow behavior (AlOmier et al., 2025). To achieve this, two distinct mixed-wettability micromodel designs were developed, enabling precise spatial control of wettability. Experimental investigations were conducted using high-resolution microscopy and high-speed imaging to capture fluid displacement dynamics. These observations were complemented by numerical simulations performed in COMSOL Multiphysics®, employing a phase-field approach for fluid interface tracking integrated with the Navier-Stokes equations. This innovative methodology allows for the creation of microfluidic systems capable of replicating complex geometries and mixed-wettability patterns. Ultimately, this study aims to provide a versatile platform for advancing the

understanding of multiphase flow in porous media, addressing critical challenges in applications such as enhanced oil recovery, CO₂ sequestration, and hydrogen storage.

Methodology

Mixed Wettability Pattern Design and Mask Writing

The microfluidic devices were fabricated using an advanced photolithography process involving two layers to precisely transfer pre-designed geometric patterns onto a substrate. This method enabled the selective allocation of wettability patterns on the flow channels, creating mixed-wettability features. The micromodel design included flow channels with a size of 100 μm and varying wettability configurations, as shown in Fig. 1a. The purple regions represented flow channels initially fabricated as hydrophilic ($\theta = 50^\circ$), while the red regions indicated areas to be modified later with a hydrophobic coating ($\theta = 113^\circ$). To investigate the effects of mixed wettability on pore-scale fluid flow behavior, two distinct wettability patterns were applied to the flow channels. The first pattern involved sequential alternating wettability, where the left half of the channel was coated with a hydrophobic layer, transitioning to a hydrophilic surface on the right half. The second pattern featured vertical variation, with the bottom half of the channel coated hydrophobically and the top half remaining hydrophilic. These configurations were designed to mimic flow through pores with heterogeneous wettability characteristics, enabling the observation of their influence on fluid displacement and flow behavior.

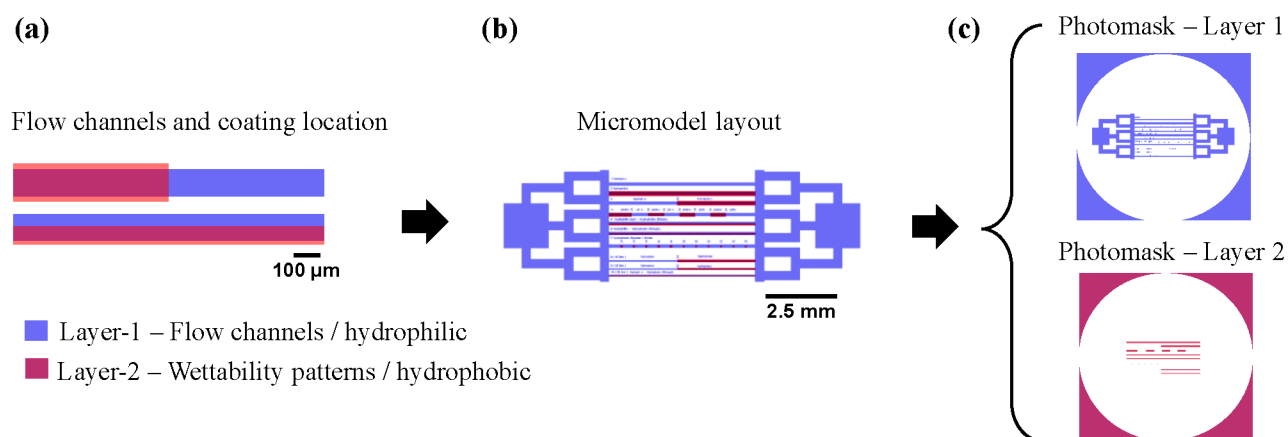


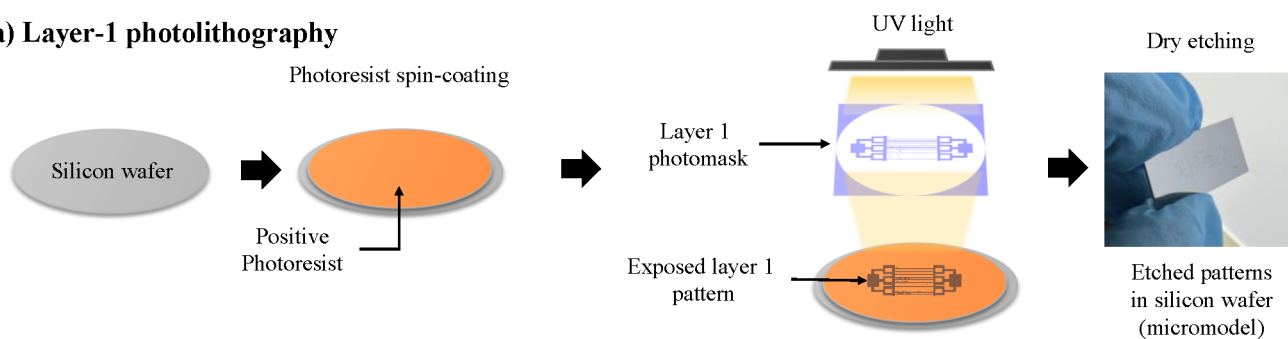
Figure 1—(a) Schematic representation of the micromodel flow channel design with mixed wettability configurations, highlighting hydrophilic (purple) and hydrophobic (red) regions. (b) Micromodel layout with dimensions of 2 cm $\times$ 1 cm, featuring repeated patterns and inlet/outlet ports for fluid injection and production. (c) Two-layer photomask design: the first layer defines the flow channels, and the second layer specifies the wettability patterns, fabricated using high-resolution laser lithography on chromium-coated glass.

These simple patterns were integrated into a micromodel layout measuring 2 cm $\times$ 1 cm, featuring inlet and outlet ports for fluid injection and production, as shown in Fig. 1b. To ensure consistent observations and minimize local flow effects, the patterns were repeated across the micromodel. The fabrication process was guided by decoupling the micromodel design into two photomask layers, illustrated in Fig. 1c. The first photomask contained the flow channel design, which was used in the first photolithography step to fabricate the micromodel structure. The second photomask included the wettability patterns to guide the coating process in the subsequent step. Both photomask layers were printed onto chromium-coated glass using a high-resolution laser lithography system. These photomasks featured selectively coated chromium areas, leaving specific regions transparent. This configuration allowed light to pass through the unmasked transparent areas while blocking it in the chromium-coated regions, enabling the precise definition of micromodel features and wettability patterns.

Mixed Wettability Microfluidics Fabrication

The workflow, depicted in Fig. 2, provides an overview of the micromodel fabrication process, emphasizing the surface modifications applied to establish mixed wettability features (AlOmier et al., 2024a). This process integrates photolithography and thin-film deposition techniques in two primary steps. In the initial step, a silicon wafer served as the base substrate for fabrication. The wafer was coated with a light-sensitive material known as positive photoresist using a spin-coating technique to ensure uniform coverage. Next, the layer-1 photomask, containing the micromodel flow pattern design, was carefully aligned over the photoresist-coated substrate. The assembly was exposed to ultraviolet (UV) light, which penetrated only the transparent regions of the photomask. This exposure rendered the UV-illuminated photoresist soluble, allowing it to be removed using a developer solution. This step left selected regions of the substrate unprotected by the photoresist, exposing them for further processing. The exposed silicon areas were then etched using a Deep Reactive-ion Etching (DRIE) technique. This process employed alternating cycles of perfluorocyclobutane (C_4F_8) and sulfur hexafluoride (SF_6) plasma to engrave the desired patterns onto the substrate with high precision. Finally, any remaining photoresist was stripped away, and the substrate was thoroughly cleaned to prepare it for subsequent steps.

(a) Layer-1 photolithography



(b) Layer-2 photolithography

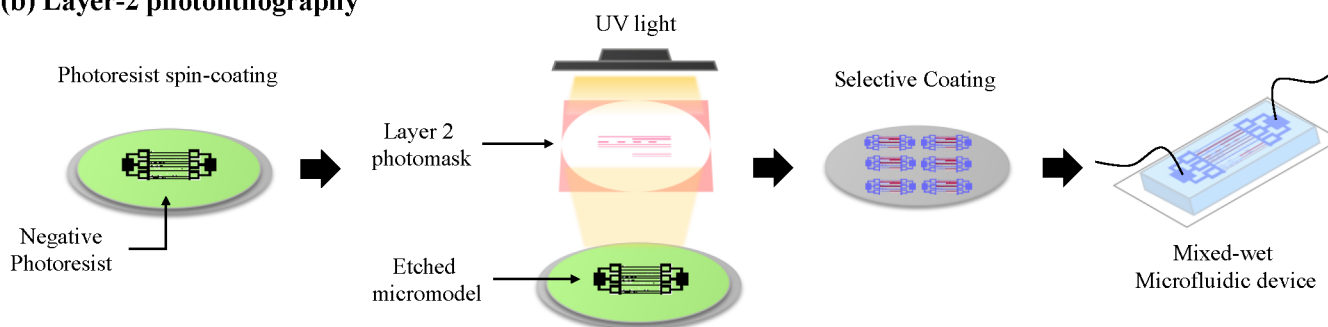


Figure 2—Workflow for fabricating mixed-wettability microfluidic devices. The process integrates layer-1 and layer-2 photolithography steps to define the flow channel patterns and selectively modify surface wettability.

In the second step, layer-2 photolithography was employed to enable selective wettability modification of the micromodel surface. The patterned silicon wafer was first spin-coated with a negative photoresist. A layer-2 photomask, containing the wettability patterns, was precisely aligned over the wafer and exposed to UV light. Upon exposure to UV light, the areas shielded by the photomask were dissolved and removed using a developing solution. This process precisely defined the targeted regions of the micromodel surface for wettability alteration, while the remaining areas were protected by the photoresist. The wafer was then subjected to hydrophobic coating with FDTs using molecular vapor deposition (MVD) in a vacuum chamber, ensuring precise and uniform coating. Following this, the remaining photoresist was lifted off using N-Methylpyrrolidone (NMP) solvent, leaving the previously protected silicon areas uncoated by

FDTs. Finally, the micromodels were sealed with a thin layer of polydimethylsiloxane (PDMS), completing the fabrication of the mixed-wet microfluidic chip.

All microfluidics flow experiments were conducted using a high-resolution optical microscope setup integrated with high-speed camera to enable direct flow observation and recording at pore-scale under dynamic conditions. Fluid injection into the microfluidic chip using highly precise syringe pump.

Mathematical and Numerical Models

The phase-field method was employed to simulate two-phase flow dynamics within the micromodels, using computational fluid dynamics (CFD) in COMSOL Multiphysics®. The phase-field function $\phi[-]$ represents the interface between the two fluids, varying from 1 to -1. The governing advection-diffusion equation is as follows (Jain, 2022):

$$\frac{\partial \phi}{\partial t} + u \nabla \phi = \nabla \cdot \left(\frac{\gamma \lambda}{\varepsilon_{pf}^2} \nabla \Psi \right), \quad (1)$$

where $u [m/s]$ is the velocity field, $\varepsilon_{pf}[m]$ controls the interface thickness, and $\gamma[m^3.s.kg^{-1}]$ and $\lambda[N]$ parameters defined as:

$$\gamma = \chi \varepsilon_{pf}^2, \quad (2)$$

$$\lambda = \frac{3 \varepsilon_{pf} \sigma}{\sqrt{8}}, \quad (3)$$

where $\chi [m.s.kg^{-1}]$ is the mobility tuning parameter, and $\sigma [N.m^{-1}]$ is the surface tension. The diffusion coefficient is given by $(\frac{\gamma \lambda}{\varepsilon_{pf}^2})$, and the chemical potential function $\Psi[-]$ is expressed as:

$$\Psi = -\nabla \cdot \left(\varepsilon_{pf}^2 \nabla \phi \right) + (\phi^2 - 1)\phi + \frac{\varepsilon_{pf}^2}{\lambda} \frac{\partial f}{\partial \phi}. \quad (4)$$

Here, $\partial f / \partial \phi [J/m^3]$ is the derivative of the free energy with respect to the phase field function. The fluid dynamics are governed by the Navier-Stokes equations for incompressible fluids, expressed as:

$$\rho u \nabla u = -\nabla p + \nabla \cdot (\mu (\nabla u + \nabla u^T)) + \rho g + F_{st}, \quad (5)$$

where $\rho [kg.m^{-3}]$ and $\mu [Pa.s]$ are the fluid density and viscosity, $p [Pa]$ is pressure, $g [m.s^{-2}]$ is gravitational acceleration (neglected due to the small model scale), and $F_{st} [N]$ is the surface tension force. At the interface, fluid properties are interpolated as:

$$\rho = \rho_1 \left(\frac{1-\phi}{2} \right) + \rho_2 \left(\frac{1+\phi}{2} \right), \quad (6)$$

$$\mu = \mu_1 \left(\frac{1-\phi}{2} \right) + \mu_2 \left(\frac{1+\phi}{2} \right). \quad (7)$$

The surface tension force is given by:

$$F_{st} = \left(\frac{\lambda}{\varepsilon_{pf}^2} \Psi - \frac{\partial f}{\partial \phi} \right) \nabla \phi. \quad (8)$$

Boundary conditions account for wettability effects, with the contact angle modeled as:

$$n \cdot \frac{\gamma \lambda}{\varepsilon_{pf}^2} \nabla \Psi = 0, \quad (9)$$

$$n \cdot \frac{\nabla \phi}{|\nabla \phi|} = \cos(\theta), \quad (10)$$

where $\theta[rd]$ is the contact angle, and n is the outward normal vector at the boundary. Equation (9) ensures zero diffusive flux across the surface, while Equation (10) specifies the contact angle. The simulations assumed incompressible, immiscible fluids under laminar, isothermal conditions, and neglect gravitational effects due to the micromodel small scale.

Results and Discussion

Experimental validation

Two-phase displacement experiments were conducted to explore the effects of mixed surface wettability, with water (dyed yellow for visualization) displacing air (white). The flow direction was maintained from left to right at a low pump flow rate, ensuring a capillary-dominated regime. Various wettability configurations were analyzed to understand the behavior of fluid interfaces across distinct wettability regions.

In the first experiment (AlOmier et al., 2025), water was injected into a single capillary channel with a wettability step transition: the left half was hydrophobic, and the right half was hydrophilic, as illustrated in Fig. 3a. During the initial stage (I), as the water advanced through the hydrophobic section, the meniscus adopted a convex shape with a dynamic contact angle of $\theta \approx 120^\circ$, characteristic of a non-wetting phase. At the interface between the hydrophobic and hydrophilic regions (stage II), the meniscus underwent an abrupt transformation, forming a nearly flat interface with a contact angle of $\theta \approx 90^\circ$. As the water progressed into the hydrophilic section (stage III), the meniscus shifted to a concave shape with a dynamic contact angle of $\theta \approx 60^\circ$, highlighting the influence of capillarity. These observations align with previous studies, which describe how wettability variations affect capillary pressure, resulting in intricate fluid displacement patterns (Lake et al., 2014).

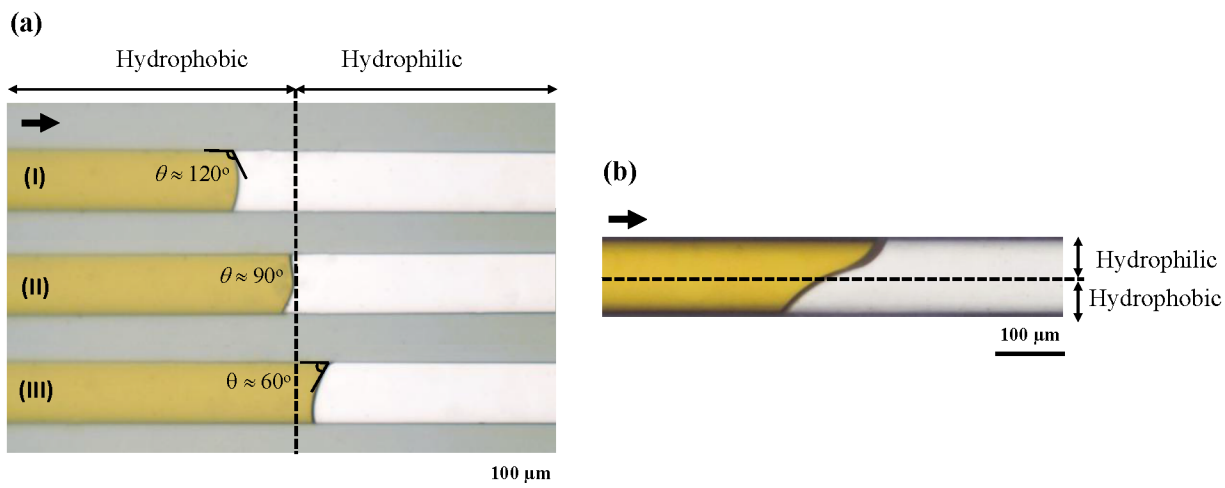


Figure 3—Experimental observations of water (yellow) displacing air (white) in microchannels with mixed wettability configurations. (a) Sequential wettability transition from hydrophobic ($\theta \approx 120^\circ$) to hydrophilic ($\theta \approx 60^\circ$) surfaces, showing distinct interface behavior at each stage: (I) hydrophobic section, (II) wettability transition, and (III) hydrophilic section. (b) S-shaped fluid interface in a channel with vertical wettability variation, exhibiting $\theta \approx 60^\circ$ against hydrophilic walls and $\theta \approx 120^\circ$ against hydrophobic walls, illustrating the impact of mixed wettability on interface curvature and capillary pressure.

In the second experiment, a capillary channel with vertical wettability distribution along its length was examined. Here, the fluid interface formed a unique S-shape along the boundary where the hydrophobic and hydrophilic surfaces meet (see Fig. 3b). The dynamic contact angles measured were $\theta \approx 60^\circ$ against the hydrophilic wall and $\theta \approx 120^\circ$ against the hydrophobic wall. This interface configuration developed as the advancing fluid moved through the wetting region while retreating within the non-wetting region, causing a pronounced bending of the fluid interface. This distinct behavior, unlike those in uniformly wetted systems,

corroborates findings by (Bauters et al., 2000; Irannezhad et al., 2023). It is attributed to the formation of near-zero mean curvatures, further complicating the estimation of capillary pressure.

Numerical Simulation

Numerical simulations were conducted to replicate the experimental conditions and validate the observed fluid behaviors. The simulated channel, measuring 860 μm in length and 100 μm in width, is illustrated in Fig. 4 (AlOmier et al., 2025). Fluid pressure and flow velocity were set as initial and boundary conditions, with the outlet pressure maintained at ambient levels, matching experimental parameters. A constant velocity boundary condition was imposed at the inlet to align with the experimental setup. Table 1 summarizes the fluid properties and parameters used in the simulations. The walls of the simulated flow domain adhered to no-slip boundary conditions.

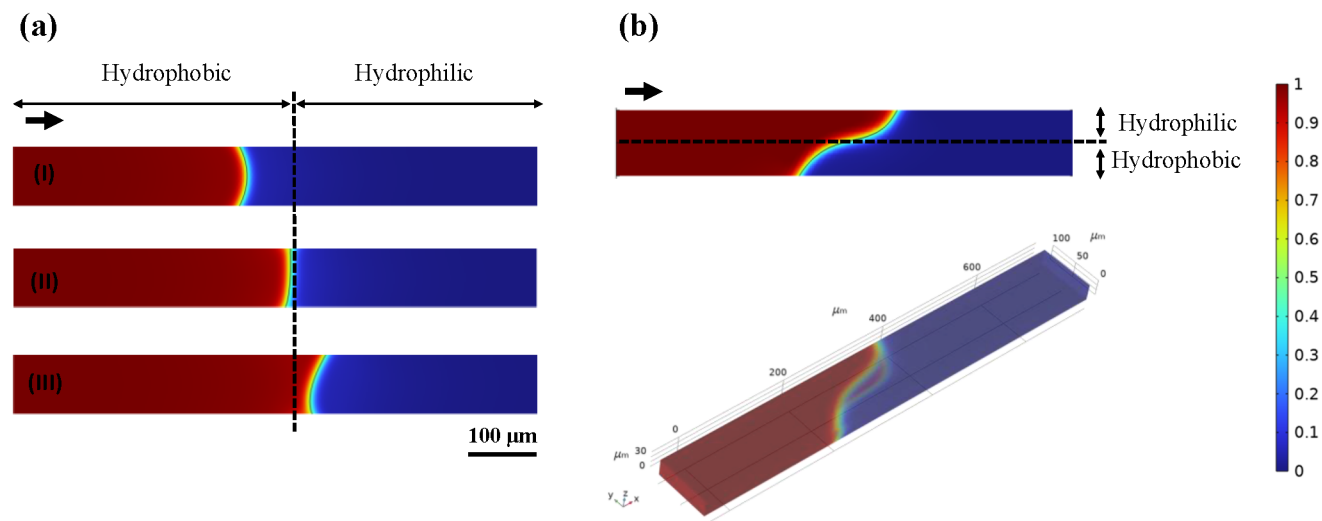


Figure 4—(a) 2D simulation of fluid displacement in a mixed-wet microchannel, with a color gradient from blue (air phase) to red (water phase) representing the fluid volume fraction during interface progression under capillary-dominated flow conditions. (b) 3D simulation of fluid behavior in a mixed-wet microchannel, highlighting the S-shaped interface and the effects of spatially variable wettability.

Table 1—Summary of fluid properties and parameters used in the numerical simulations for replicating two-phase flow behavior in mixed-wettability microfluidic channels.

Parameter	Value
$\rho_1[\text{kg.m}^{-3}]$	1.2032
$\rho_2[\text{kg.m}^{-3}]$	998.06
$\mu_1[\text{Pa.s}]$	1.81×10^{-5}
$\mu_2[\text{Pa.s}]$	9.99×10^{-4}
$\varepsilon_{pf}[\text{m}]$	6.5×10^{-6}
$\chi [\text{m.s.kg}^{-1}]$	1
$\sigma [\text{N.m}^{-1}]$	0.0726
$\partial f / \partial \phi [\text{J/m}^3]$	0

The phase-field model incorporated surface wettability by assigning contact angles as boundary conditions, with $\theta = 120^\circ$ for hydrophobic surfaces and $\theta = 60^\circ$ for hydrophilic surfaces. A computational mesh of approximately 2700 nodes was generated using COMSOL Multiphysics®. The progression of fluid front within the domain is depicted in Fig. 4a. The simulation results use a color gradient—blue representing

the air phase and red representing the water phase—to depict the displacing volume fraction of the fluid. These numerical simulations effectively captured the fluid interface behavior and shape, showing strong agreement with the experimental results.

The second case was modeled in three dimensions (3D) to account for the spatial variability of the boundary conditions, as illustrated in Fig. 4b. The simulated domain had a length of 730 μm , with a computational mesh comprising approximately 15,500 nodes to ensure adequate resolution for capturing detailed fluid dynamics. Surface wettability was incorporated into the boundary walls using the phase-field model, replicating experimental conditions with contact angles of $\theta = 120^\circ$ for hydrophobic surfaces and $\theta = 60^\circ$ for hydrophilic surfaces. All other simulation parameters remained consistent with those applied in the first case. The simulations successfully replicated the distinctive S-shaped interface observed during water displacement in the mixed-wet channel. The 3D model effectively demonstrated the bending of the fluid interface caused by the hydrophobic top layer sealing the channel, leading to more pronounced spreading along the channel hydrophilic bottom surface. This behavior mirrored experimental results, where a thicker interface was visible in the hydrophilic section when viewed from above. These consistent findings affirm the reliability of the simulation model in accurately capturing the influence of mixed wettability on fluid flow dynamics.

Conclusions

This study demonstrated the successful fabrication and experimental validation of mixed-wettability microfluidic devices using advanced photolithography and thin-film deposition techniques. By designing and implementing specific wettability patterns, the study investigated the impact of heterogeneous surface properties on two-phase fluid dynamics. Experimental results revealed distinct interface behaviors influenced by mixed-wet configurations, including wettability-induced interface transitions and S-shaped interface profiles. Numerical simulations, using the phase-field approach, accurately replicated these experimental observations, confirming the validity of the mathematical models in capturing the effects of mixed wettability on fluid displacement. The integration of experimental and numerical methodologies provides a robust platform for understanding complex multiphase flow dynamics in porous media, with significant implications for applications in enhanced oil recovery, CO₂ sequestration, and other subsurface technologies.

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